

A Bi-Level Decomposition Matheuristic with Adaptive Operator Selection for the Multi-Objective Movie Scene Scheduling Problem

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Abstract

The Movie Scene Scheduling Problem (MSSP) seeks to sequence film scenes across shooting days to minimize total production cost and makespan, subject to a rich set of real-world constraints including actor availability, hierarchical location logistics, crew travel continuity, labor regulations, and deadline penalties. Existing approaches either solve simplified MSSP variants exactly via mixed-integer linear programming (MILP), which becomes intractable beyond 15–20 scenes, or apply standalone meta-heuristics that ignore exploitable problem structure. We first present a comprehensive MILP formulation with **13 constraint families** over a three-tier geographic hierarchy (city–town–location), actor–scene–day and location–scene–day linking variables, town-uniqueness and town-continuity logistics constraints, criticality-weighted utilization, and consecutive working-day limits—substantially extending prior formulations. We then propose **BDM-AOS**, a Bi-level Decomposition Matheuristic with Adaptive Operator Selection. Our method introduces: (i) a *Scene Interaction Graph* (SIG) that captures actor-sharing, location-sharing, and transfer-cost coupling between scenes, enabling Louvain community detection to partition scenes into tightly coupled *shooting blocks*; (ii) a bi-level framework where NSGA-II evolves block permutations at the upper level while a global greedy decoder schedules scenes in block-guided order, simultaneously optimizing cost and makespan; (iii) a Q-learning adaptive operator selection mechanism that dynamically chooses among standard and two novel problem-specific operators—Location-Aware Crossover (LAX) and Actor-Continuity Mutation (ACM). Comprehensive experiments on 8 synthetic instance sizes (10–100 scenes) with 30 independent seeds per algorithm reveal a scalability crossover: while GA and PSO outperform BDM-AOS on small instances (≤ 30 scenes), BDM-AOS achieves **20.7% lower cost** than GA and **19.0% lower cost** than PSO on 75-scene instances, with a **5.6 $\times$ runtime speedup**. On the largest 100-scene instances, BDM-AOS maintains a **14.5%** cost advantage over GA. A five-variant ablation study identifies SIG decomposition as the critical component (33–64% degradation without it). Evaluation uses hypervolume (HV) and inverted generational distance (IGD) quality indicators, with Friedman tests and Holm’s post-hoc correction for statistical rigor. All code and data are publicly available at <https://github.com/VJlaxmi/movie-scheduling-paper>.

1 Introduction

Film production is a multi-million dollar industry where shooting schedules directly impact budgets. The *Movie Scene Scheduling Problem* (MSSP) determines the order and day assignments of scenes to minimize total production cost—comprising actor wages, hierarchical location transfer costs, deadline penalties, criticality-weighted resource utilization, and actor holding fees—subject to a rich set of real-world constraints: actor availability calendars, a three-tier geographic location hierarchy (city–town–location), crew travel logistics that restrict town changes across consecutive days, daily shooting capacity, precedence relations, time windows, labor regulations limiting consecutive working days, specialized equipment sharing, mandatory scene co-scheduling groups, minimum scene separation requirements, location concurrency capacities, and lead-actor rest period requirements.

The MSSP is NP-hard, reducing to a variant of the resource-constrained project scheduling problem (RCPSP) [4]. Exact methods based on integer linear programming (ILP) can find provably optimal solutions for small instances (10–15 scenes) but become intractable as problem size grows [29, 39]. Metaheuristic approaches—including tabu search [29], particle swarm optimization [30, 39], and ant colony optimization [30]—offer scalable alternatives but operate on flat scene permutations, ignoring the structural coupling between scenes that share actors or locations. Moreover, existing formulations employ simplified constraint models that omit critical production logistics such as hierarchical location structures, town continuity requirements, and actor labor regulations.

We observe that real film schedules exhibit *shooting blocks*: groups of scenes that are tightly coupled through shared actors and co-located sets, and are typically filmed consecutively in practice. This observation motivates a decomposition approach that exploits this structure.

1.1 Contributions

We propose **BDM-AOS** (Bi-level Decomposition Matheuristic with Adaptive Operator Selection), which makes the following contributions:

1. **Comprehensive MILP formulation:** Building on our prior single-objective scheduling work [27], we present the most complete MSSP formulation to date, with **19 constraint families** over a three-tier geographic hierarchy (city–town–location): the 13-family baseline extended with equipment sharing (C14), actor booking/holding fees (C15), mandatory co-scheduling (C16), minimum scene separation (C17), location concurrency (C18), and lead-actor rest periods (C19). This substantially extends prior formulations [29, 39] with operational constraints absent from all existing models.
2. **Scene Interaction Graph (SIG):** A novel graph-theoretic representation where nodes are scenes and edge weights quantify coupling through actor sharing (Jaccard similarity), location sharing, and transfer cost proximity. Louvain community detection [2] on the SIG partitions scenes into shooting blocks, dramatically reducing the combinatorial search space.
3. **Bi-level Decomposition with Greedy Reassembly:** The upper level uses NSGA-II [10] to evolve permutations of shooting blocks, simultaneously optimizing total cost and makespan. The lower level applies a global greedy decoder that packs scenes into days in block-guided order, enforcing all 19 constraint families. This is the first decomposition-based approach for the MSSP.
4. **Q-Learning Adaptive Operator Selection (AOS):** A reinforcement learning mechanism that dynamically selects among a pool of six crossover-mutation combinations, including two novel problem-specific operators:
 - *Location-Aware Crossover (LAX)*: preserves sub-sequences where consecutive blocks share the same primary location;

- *Actor-Continuity Mutation (ACM)*: reduces actor idle gaps by repositioning blocks to improve actor schedule continuity.
5. **Multi-objective formulation**: We present the first bi-objective MSSP formulation optimizing cost and makespan simultaneously, providing a Pareto front of trade-off solutions.
 6. **Experimental evaluation**: Comprehensive experiments on 8 instance sizes (10–100 scenes) with 30 independent seeds per algorithm, five ablation variants, hypervolume and inverted generational distance quality indicators, and Friedman tests with Holm’s post-hoc correction [11, 15, 21]. We include a detailed MILP-vs.-BDM-AOS comparison (Section 4.3) reporting both objective costs and runtimes, showing BDM-AOS matches MILP optimality on small instances with up to $186\times$ speedup, and remains the only feasible approach for $n \geq 30$.

The remainder of this paper is organized as follows. Section 2 reviews related work. Section 3 presents the problem formulation and the BDM-AOS algorithm. Section 4 describes the experimental setup and results. Section 5 discusses findings and implications. Section 6 addresses limitations. Section 7 concludes with future research directions.

2 Related Work

We survey five bodies of literature that inform the design of BDM-AOS: (i) exact methods for the MSSP, (ii) metaheuristic approaches to scene scheduling, (iii) multi-objective evolutionary optimization, (iv) decomposition-based optimization and community detection, and (v) adaptive operator selection and reinforcement learning.

2.1 Exact Methods for the MSSP

The MSSP was first formalized as a combinatorial optimization problem with integer linear programming [3, 7]. Zhen et al. [43] formulated a cost-minimization ILP and solved it with CPLEX. Liu et al. [29] extended this model with 24 constraints covering actor availability, location compatibility, and scene-day assignments, solving instances up to 15 scenes optimally before CPLEX exceeded 7,200-second time limits. Tekin et al. [39] developed a comprehensive MILP formulation with 31 constraints including precedence relations, time windows, and actor blocking periods. Using CPLEX on 160 benchmark instances, Tekin showed that exact solutions become impractical beyond 25 scenes with blocking-time constraints. This NP-hard complexity profile [16] motivates metaheuristic and matheuristic approaches.

Notably, all existing MILP formulations for the MSSP employ *flat* location models—locations are treated as independent entities without geographic structure. Real film productions operate across a multi-tier hierarchy (production cities, towns/districts, and specific venues), with transfer costs and crew logistics that vary by tier. Our prior work [27] introduced a 13-family formulation with the three-tier location hierarchy and bi-objective treatment. The present paper extends it to a **19-family formulation** by adding equipment sharing, actor booking contracts with holding fees, mandatory co-scheduling groups, minimum scene separation, location concurrency limits, and lead-actor rest periods—constraints present in real production contracts [3] but absent from all prior computational models.

2.2 Metaheuristic Approaches

Liu et al. [29] introduced a Tabu Search-based metaheuristic (TSBM) achieving average gaps of 0.18% from optimal on instances with 10–40 scenes and 3–20 actors, along with a PSO-based method (PSOBM) with 7.92% average gaps. Long et al. [30] added an Ant Colony Optimization

method (ACOBM) and showed ACO outperformed both TSBM and PSOBM in solution quality on larger instances, paralleling results from broader swarm intelligence literature [5, 12, 25]. Tekin et al. [39] developed a discrete PSO with binary position representation and feasibility-based repair, achieving less than 1% deviation from optimal on 106 out of 160 instances and finding better solutions than CPLEX (which timed out) on a real film script with 70 scenes and 43 actors.

All existing metaheuristics operate on flat scene permutations without exploiting the coupling structure between scenes. None employ multi-objective optimization, decomposition strategies, or adaptive operator selection. Genetic algorithm operators for permutation-based problems are well-studied [34, 38, 41], but no MSSP-specific crossover or mutation operators have been proposed prior to this work.

2.3 Multi-Objective Evolutionary Optimization

Multi-objective evolutionary algorithms (MOEAs) have been widely applied to scheduling problems [6, 8]. Deb et al. [10] introduced NSGA-II, which remains the standard for bi-objective optimization; Deb and Jain [9] extended it to many-objective problems via reference-point-based sorting (NSGA-III). In a related film-production context, Han et al. [19] applied NSGA-III to visual effects composition—a distinct problem from scene scheduling—using Wilcoxon and Friedman tests for algorithm comparison.

Performance assessment of MOEAs relies on quality indicators such as the hypervolume indicator [22, 45] and inverted generational distance [1, 44]. These metrics enable rigorous comparison of Pareto fronts across algorithms. We adopt both indicators in our experimental evaluation (Section 4).

2.4 Decomposition and Community Detection

Decomposition approaches partition large optimization problems into coordinated sub-problems. Zhang and Li [42] introduced MOEA/D, a decomposition-based MOEA that has become a leading paradigm. Graph-based decomposition exploits problem structure through community detection algorithms [14, 17], with the Louvain method [2] being particularly effective for modularity maximization. General decomposition frameworks for large-scale optimization [36] demonstrate that structure-aware partitioning consistently outperforms random decomposition.

Matheuristics—hybrids of mathematical programming and metaheuristics—have shown strong performance on combinatorial optimization [23, 31]. Decomposition-based matheuristics solve sub-problems exactly while using metaheuristics for global coordination. Resource-constrained project scheduling [20, 26, 35] provides the closest methodological parallel, though film scheduling introduces unique constraints (actor availability, location dependencies, cost structures) not present in standard RCPSP formulations. To our knowledge, no matheuristic has been applied to the MSSP.

2.5 Adaptive Operator Selection and Reinforcement Learning

Adaptive operator selection (AOS) uses online learning to choose effective operators during search [24]. Maturana et al. [32] proposed autonomous operator management for EAs, showing that adaptive selection consistently outperforms fixed operator strategies. Biased random-key GAs [18] demonstrate how problem-specific encodings improve search performance.

Q-learning-based AOS [37, 40] adapts operator probabilities based on search state features—population diversity, improvement rate, and convergence stage. This state-action framework enables the search process to learn which operators are most effective in different search contexts, as demonstrated by Li et al. [28] and Fialho et al. [13]. Statistical testing protocols for comparing stochastic algorithms [11, 15] require non-parametric methods such as Friedman tests with Holm’s post-hoc correction [21].

Table 1: Positioning of BDM-AOS relative to existing MSSP approaches.

Feature	Liu (2019)	Long (2020)	Tekin (2023)	BDM-AOS (Ours)
MILP constraint families	24	—	31	19 (structured)
Location hierarchy	flat	flat	flat	3-tier (C/T/L)
Town logistics constraints	—	—	—	C10–C12
Actor labor + rest	—	—	blocking	C13, C19
Equipment sharing	—	—	—	C14
Actor booking / holding fee	—	—	—	C15
Scene coordination	—	—	—	C16–C17
Location concurrency	—	—	—	C18
Criticality weighting	—	—	—	✓
Single metaheuristic	TS, PSO	TS, PSO, ACO	PSO	—
Decomposition	—	—	—	SIG+Louvain
Bi-level framework	—	—	—	✓
Multi-objective	—	—	—	NSGA-II
Adaptive operator select.	—	—	—	Q-learning
Problem-specific operators	—	—	—	LAX, ACM
Max scenes	40	40	70	100
Independent seeds	1	1	1	30
Statistical testing	—	—	—	Friedman+Holm
Quality indicators	Gap%	Gap%	Gap%	HV, IGD

2.6 Positioning of This Work

Table 1 summarizes the landscape. BDM-AOS is the first MSSP approach combining (i) the richest MILP formulation to date with 19 structured constraint families over a three-tier location hierarchy, including equipment sharing, actor booking contracts, mandatory scene coordination, and lead-actor welfare constraints, (ii) decomposition via graph clustering, (iii) a bi-level metaheuristic-decoder framework enforcing all 19 constraint families, (iv) multi-objective optimization, and (v) adaptive operator selection with problem-specific operators.

3 Methodology

3.1 Problem Formulation

We formulate the MSSP as a multi-objective mixed-integer program over a hierarchical location structure, capturing the full complexity of real-world film production logistics. Building on our earlier single-objective scheduling work [27], we substantially extend the constraint model with six new constraint families governing equipment sharing, actor booking contracts, mandatory scene coordination, location concurrency, and lead-actor welfare regulations—yielding a formulation with **19 constraint families** that addresses operational dimensions absent from prior work [29, 39].

Primary Sets. Let $\mathcal{S} = \{s_1, \dots, s_n\}$ be the set of scenes, $\mathcal{A} = \{a_1, \dots, a_m\}$ the actors, $\mathcal{D} = \{1, \dots, T\}$ the planning horizon of available shooting days, and $\mathcal{P} \subseteq \mathcal{S} \times \mathcal{S}$ the set of precedence pairs. Let $\mathcal{A}^P \subseteq \mathcal{A}$ denote the subset of *principal (lead) actors* subject to enhanced welfare constraints.

Hierarchical Location Model. Real film productions operate across a three-tier geographic hierarchy: *cities* (production bases), *towns* (districts within cities), and *locations* (specific shooting venues within towns). We model this via three sets and two surjective mappings:

- $\mathcal{C} = \{c_1, \dots, c_p\}$: cities (production hubs);
- $\mathcal{I} = \{i_1, \dots, i_q\}$: towns, with $\mathcal{I}_c \subseteq \mathcal{I}$ denoting the towns in city c ;
- $\mathcal{L} = \{l_1, \dots, l_k\}$: locations, with $\mathcal{L}_i \subseteq \mathcal{L}$ denoting the locations in town i ;
- $\phi : \mathcal{L} \rightarrow \mathcal{I}$: maps each location to its town, $\phi(l) = i \iff l \in \mathcal{L}_i$;
- $\psi : \mathcal{I} \rightarrow \mathcal{C}$: maps each town to its city, $\psi(i) = c \iff i \in \mathcal{I}_c$.

This induces nested containment: $l \in \mathcal{L}_i \Rightarrow i \in \mathcal{I}_{\psi(i)}$, ensuring that activating a location implies its town and city are also active.

Scene Parameters. Each scene $s \in \mathcal{S}$ has: shooting duration δ_s (hours); required actors $A(s) \subseteq \mathcal{A}$; compatible locations $L(s) \subseteq \mathcal{L}$; optional time window $[e_s, l_s] \subseteq \mathcal{D}$; optional soft deadline $\bar{d}_s$ with per-day penalty ρ_s .

Actor Parameters. Each actor $a \in \mathcal{A}$ has: daily wage w_a ; availability calendar $\mathcal{D}_a \subseteq \mathcal{D}$; criticality weight $\kappa_a^A \in \mathbb{R}_{>0}$ (reflecting contractual priority); maximum consecutive working days $\bar{h}_a$ (labor regulations); booking window $[\beta_a^s, \beta_a^e] \subseteq \mathcal{D}$ (period under production contract); daily holding fee $h_a \geq 0$ (payment when booked but not shooting); minimum post-block rest duration r_a (applicable to $a \in \mathcal{A}^P$).

Equipment Parameters. Let $\mathcal{E} = \{e_1, \dots, e_f\}$ be the set of specialized equipment types (cameras, lighting rigs, cranes, etc.). Each scene s requires equipment subset $E(s) \subseteq \mathcal{E}$, and the production has $\xi_e \in \mathbb{Z}_{>0}$ units of equipment type e available.

Location and Transfer Parameters. Each location $l \in \mathcal{L}$ has: availability calendar $\mathcal{D}_l \subseteq \mathcal{D}$; criticality weight $\kappa_l^L \in \mathbb{R}_{>0}$ (reflecting booking difficulty); concurrency capacity $\mu_l \in \mathbb{Z}_{>0}$ (maximum simultaneous scenes at location l per day). Transfer costs capture three tiers: intra-town $\tau_{l,l'}^{\text{loc}}$, inter-town $\tau_{i,i'}^{\text{town}}$, inter-city $\tau_{c,c'}^{\text{city}}$. The composite transfer cost is:

$$c_{l,l'} = \begin{cases} \tau_{l,l'}^{\text{loc}} & \text{if } \phi(l) = \phi(l') \\ \tau_{\phi(l),\phi(l')}^{\text{town}} + \tau_{l,l'}^{\text{loc}} & \text{if } \psi(\phi(l)) = \psi(\phi(l')), \phi(l) \neq \phi(l') \\ \tau_{\psi(\phi(l)),\psi(\phi(l'))}^{\text{city}} + \tau_{\phi(l),\phi(l')}^{\text{town}} + \tau_{l,l'}^{\text{loc}} & \text{otherwise} \end{cases} \quad (1)$$

Scene Coordination Parameters. Let $\mathcal{G} = \{G_1, \dots, G_p\}$ be a collection of *mandatory co-scheduling groups*: for each $G_k \in \mathcal{G}$, all scenes in G_k must be shot on the same day (e.g., narrative continuity shots, special-effects setups requiring joint preparation). Let $\mathcal{F} \subseteq \mathcal{S} \times \mathcal{S}$ be the set of *separated scene pairs* with minimum required inter-scene gaps $\delta_{ij} \geq 1$ (e.g., scenes (s_i, s_j) requiring δ_{ij} days between them for visual continuity, cast transformation, or location reset).

Binary Indicator Parameters. For notational convenience we define: $b_{a,s} = \mathbb{K}[a \in A(s)]$ (actor a required for scene s); $\alpha_{a,d} = \mathbb{K}[d \in \mathcal{D}_a]$ (actor a available on day d); $\lambda_{l,d} = \mathbb{K}[d \in \mathcal{D}_l]$ (location l available on day d); $p_{s,l} = \mathbb{K}[l \in L(s)]$ (scene s compatible with location l); $\eta_{i,l} = \mathbb{K}[l \in \mathcal{L}_i]$ (location l belongs to town i); $\mu_{a,d}^b = \mathbb{K}[d \in [\beta_a^s, \beta_a^e]]$ (actor a under contract on day d); $\varepsilon_{e,s} = \mathbb{K}[e \in E(s)]$ (scene s requires equipment type e).

Decision Variables. The model uses seven families of binary decision variables:

$$x_{s,d} \in \{0, 1\} \quad \text{scene } s \text{ scheduled on day } d \quad (2)$$

$$y_{s,l} \in \{0, 1\} \quad \text{scene } s \text{ assigned to location } l \quad (3)$$

$$v_{a,s,d} \in \{0, 1\} \quad \text{actor } a \text{ participates in scene } s \text{ on day } d \quad (4)$$

$$z_{l,s,d} \in \{0, 1\} \quad \text{scene } s \text{ shot at location } l \text{ on day } d \quad (5)$$

$$u_{i,d} \in \{0, 1\} \quad \text{town } i \text{ is the active shooting town on day } d \quad (6)$$

$$\phi_{a,d} \in \{0, 1\} \quad \text{actor } a \text{ is idle (non-shooting) on booked day } d \quad (7)$$

$$w_{a,d} \in \{0, 1\} \quad \text{actor } a \text{ works (in at least one scene) on day } d \quad (8)$$

The linking variables $v_{a,s,d}$ and $z_{l,s,d}$ couple scene scheduling with actor participation and location usage. The town activation variable $u_{i,d}$ enforces geographic crew logistics. The booking idle variable $\phi_{a,d}$ captures contracted days on which an actor is not shooting, triggering holding fees in the objective. The working-day indicator $w_{a,d}$ provides a scene-agnostic view of actor activity, enabling enforcement of lead-actor rest periods across variable-length shooting windows.

Bi-Objective Formulation. We minimize two conflicting objectives simultaneously:

$$\begin{aligned} f_1(\mathbf{x}, \mathbf{v}, \mathbf{z}, \phi) = & \underbrace{\sum_{a \in \mathcal{A}} w_a \cdot |\{d \in \mathcal{D} : \sum_s v_{a,s,d} \geq 1\}|}_{\text{(i) actor daily wages}} + \underbrace{\sum_{d=2}^T \sum_{\substack{l, l' \in \mathcal{L} \\ l \neq l'}} c_{l,l'} \cdot \sigma_{l,l',d}}_{\text{(ii) hierarchical transfer costs}} \\ & + \underbrace{\sum_{s \in \mathcal{S}} \rho_s \cdot \max(0, \sum_d d \cdot x_{s,d} - \bar{d}_s)}_{\text{(iii) deadline penalties}} + \underbrace{\sum_{(l,s,d)} \frac{z_{l,s,d}}{\kappa_l^L} + \sum_{(a,s,d)} \frac{v_{a,s,d}}{\kappa_a^A}}_{\text{(iv) criticality-weighted utilization}} \\ & + \underbrace{\sum_{a \in \mathcal{A}} h_a \cdot \sum_{d \in \mathcal{D}} \phi_{a,d}}_{\text{(v) actor holding fees}} \end{aligned} \quad (9)$$

$$f_2(\mathbf{x}) = \max_{s \in \mathcal{S}} \sum_d d \cdot x_{s,d} - \min_{s \in \mathcal{S}} \sum_d d \cdot x_{s,d} + 1 \quad (10)$$

where $\sigma_{l,l',d} = 1$ if location l' is used on day d and location l was used on day $d-1$. Cost objective f_1 has five terms: (i) actor wages paid per active day, (ii) hierarchical transfer costs from Eq. (1), (iii) deadline penalties, (iv) criticality-weighted utilization penalizing usage of low-criticality resources, and (v) actor holding fees paid when a contracted actor is not shooting. Makespan f_2 minimizes total production duration.

Constraints. The formulation enforces **19 constraint families** organized into eight categories. After each group of formal constraints, we provide a brief real-world interpretation to clarify the operational meaning.

I. Assignment Constraints.

$$\sum_{d \in \mathcal{D}} x_{s,d} = 1 \quad \forall s \in \mathcal{S} \quad (C1)$$

$$\sum_{l \in L(s)} y_{s,l} = 1 \quad \forall s \in \mathcal{S} \quad (C2)$$

$$\text{day}(s_i) + 1 \leq \text{day}(s_j) \quad \forall (s_i, s_j) \in \mathcal{P} \quad (C3)$$

$$e_s \leq \text{day}(s) \leq l_s \quad \forall s \text{ with time windows} \quad (C4)$$

Interpretation: **C1** schedules each scene exactly once—no scene may be split, deferred indefinitely, or duplicated. **C2** assigns each scene to exactly one compatible location—a scene cannot be simultaneously staged at multiple venues. **C3** enforces narrative or logistic precedence with at least a one-day separation; e.g., a scene set before dawn must precede its sequel. **C4** restricts scene scheduling to its permitted shooting window (e.g., seasonal or contractual release windows), where $\text{day}(s) = \sum_d d \cdot x_{s,d}$.

II. Capacity and Availability Constraints.

$$\sum_{s \in \mathcal{S}} \delta_s \cdot x_{s,d} \leq W_{\max} \quad \forall d \in \mathcal{D} \quad (\text{C5})$$

$$x_{s,d} \leq \alpha_{a,d} \quad \forall s \in \mathcal{S}, a \in A(s), d \in \mathcal{D} \quad (\text{C6})$$

$$z_{l,s,d} \leq \lambda_{l,d} \quad \forall l \in L(s), s \in \mathcal{S}, d \in \mathcal{D} \quad (\text{C7})$$

Interpretation: **C5** caps the total shooting duration per calendar day at $W_{\max} = 8$ hours, reflecting union-mandated shooting limits and crew fatigue regulations. **C6** prevents scheduling a scene on any day when at least one required actor is unavailable (due to illness, other commitments, or contractual non-availability). **C7** ensures locations are used only on days when they are accessible—a location may be closed for maintenance, reserved by other productions, or subject to permit restrictions.

III. Variable Linking Constraints.

$$v_{a,s,d} = x_{s,d} \cdot b_{a,s} \cdot \alpha_{a,d} \quad \forall a \in \mathcal{A}, s \in \mathcal{S}, d \in \mathcal{D} \quad (\text{C8})$$

$$z_{l,s,d} = x_{s,d} \cdot p_{s,l} \cdot \lambda_{l,d} \quad \forall l \in \mathcal{L}, s \in \mathcal{S}, d \in \mathcal{D} \quad (\text{C9})$$

$$w_{a,d} \geq v_{a,s,d} \quad \forall a \in \mathcal{A}, s \in \mathcal{S}, d \in \mathcal{D} \quad (\text{C9b})$$

$$w_{a,d} \leq \sum_{s \in \mathcal{S}} v_{a,s,d} \quad \forall a \in \mathcal{A}, d \in \mathcal{D} \quad (\text{C9c})$$

Interpretation: **C8** forces $v_{a,s,d} = 1$ exactly when scene s is on day d , actor a is in the cast of s , and a is available—providing the fine-grained actor-day participation record needed for wage calculation and labor-limit enforcement. **C9** forces $z_{l,s,d} = 1$ exactly when scene s is shot at location l on day d and l is available—enabling transfer cost computation across the three-tier hierarchy. Since $b_{a,s}$, $\alpha_{a,d}$, $p_{s,l}$, $\lambda_{l,d}$ are fixed binary parameters, C8 and C9 reduce to linear equalities. **C9b–C9c** together define the *working-day indicator* $w_{a,d}$: $w_{a,d} = 1$ if and only if actor a appears in at least one scene on day d . This scene-agnostic view of actor activity is required by both the consecutive-day labor limit **C13** and the lead-actor rest constraint **C19**.

IV. Geographic Logistics Constraints.

$$z_{l,s,d} \leq u_{\phi(l),d} \cdot \eta_{\phi(l),l} \cdot \lambda_{l,d} \quad \forall l \in L(s), s \in \mathcal{S}, d \in \mathcal{D} \quad (\text{C10})$$

$$\sum_{i \in \mathcal{I}} u_{i,d} \leq 1 \quad \forall d \in \mathcal{D} \quad (\text{C11})$$

$$u_{i,d} + u_{j,d-1} \leq 1 \quad \forall i, j \in \mathcal{I}, i \neq j, d \geq 2 \quad (\text{C12})$$

$$\sum_{d'=d}^{\min(d+\bar{h}_a-1, T)} w_{a,d'} \leq \bar{h}_a \quad \forall a \in \mathcal{A}, d \in \mathcal{D} \quad (\text{C13})$$

Interpretation: **C10** enforces location–town consistency: using location l on day d requires its town $\phi(l)$ to be the active town. **C11** restricts the crew to *at most one active town per day*—crew and equipment cannot operate in multiple geographic areas simultaneously; all active scenes must be within the same town district. **C12** enforces *town continuity across consecutive days*: the crew cannot switch towns between successive shooting days without a dedicated travel day, capturing real-world overnight relocation infeasibility. **C13** limits each actor to at most $\bar{h}_a$ *working days* (via the indicator $w_{a,d'}$ from **C9b–C9c**) in any rolling window of $\bar{h}_a$ consecutive calendar days,

modeling union-mandated limits on consecutive working days and preventing crew burnout. Using $w_{a,d'}$ rather than $\sum_s v_{a,s,d'}$ is essential: an actor appearing in multiple scenes on the same day counts as one working day, not multiple, correctly tracking the day-level labor constraint.

V. Equipment Sharing Constraints.

$$\sum_{s \in \mathcal{S}} \varepsilon_{e,s} \cdot x_{s,d} \leq \xi_e \quad \forall e \in \mathcal{E}, d \in \mathcal{D} \quad (\text{C14})$$

Interpretation: C14 enforces that on any given day, the total number of scenes simultaneously requiring equipment type e (cranes, underwater rigs, high-speed cameras, etc.) cannot exceed the available supply ξ_e . This models a critical real-world production bottleneck: specialized equipment is expensive to rent and often available in limited quantities, forcing the scheduler to spread equipment-intensive scenes across days. For standard equipment with $\xi_e \geq n$, this constraint is non-binding and incurs no overhead.

VI. Actor Booking and Holding Fee Constraints.

$$\phi_{a,d} \geq \mu_{a,d}^b - \sum_{s \in \mathcal{S}} v_{a,s,d} \quad \forall a \in \mathcal{A}, d \in \mathcal{D} \quad (\text{C15})$$

$$\phi_{a,d} \leq \mu_{a,d}^b \quad \forall a \in \mathcal{A}, d \in \mathcal{D} \quad (\text{C15b})$$

Interpretation: C15 and C15b together force $\phi_{a,d} = 1$ exactly when actor a is under contract on day d (i.e., $\mu_{a,d}^b = 1$) but not shooting any scene. This models the real-world *pay-or-play* clause in actor contracts: a production that books an actor for a window $[\beta_a^s, \beta_a^e]$ must pay the holding fee h_a for every idle day within that window. The incentive created is to schedule scenes for contracted actors promptly within their booking windows, avoiding unnecessary idle days and wasted holding payments.

VII. Scene Coordination Constraints.

$$x_{s,d} = x_{s',d} \quad \forall s, s' \in G_k, G_k \in \mathcal{G}, d \in \mathcal{D} \quad (\text{C16})$$

$$\sum_{d \in \mathcal{D}} d \cdot x_{s_j,d} - \sum_{d \in \mathcal{D}} d \cdot x_{s_i,d} \geq \delta_{ij} \quad \forall (s_i, s_j) \in \mathcal{F} \quad (\text{C17})$$

Interpretation: C16 enforces mandatory co-scheduling: all scenes within group G_k must be shot on the same calendar day. This arises when a single elaborate set or lighting arrangement is used by multiple scenes (e.g., a complex ballroom scene requires the same setup for several shots that cannot be split across days without incurring full re-setup costs), or when scenes share a one-take practical effect. C17 enforces minimum separation between paired scenes: scene s_j must be scheduled at least δ_{ij} days after scene s_i . Real use cases include: post-production visual-effects windows (an on-set scene must be completed before a VFX plate is shot), actor transformation time (scenes requiring different hair/makeup states), and location reset time (a scene requiring demolition or irreversible set changes before a sequel scene can be filmed).

VIII. Location Concurrency and Lead-Actor Rest Constraints.

$$\sum_{s \in \mathcal{S}} z_{l,s,d} \leq \mu_l \quad \forall l \in \mathcal{L}, d \in \mathcal{D} \quad (\text{C18})$$

$$\sum_{d'=d}^{\min(d+\bar{h}_a+r_a-1, T)} w_{a,d'} \leq \bar{h}_a \quad \forall a \in \mathcal{A}^P, d \in \mathcal{D} \quad (\text{C19})$$

Interpretation: C18 limits the number of scenes simultaneously active at location l on any single day to μ_l . Some locations physically cannot support more than μ_l simultaneous shooting units due to space constraints, permit conditions, or acoustic interference between adjacent scenes (e.g., $\mu_l = 1$ for a small interior, $\mu_l = 3$ for a large studio with multiple sound stages). C19 enforces a mandatory

rest window for principal actors: within any rolling window of $\bar{h}_a + r_a$ consecutive days, a lead actor $a \in \mathcal{A}^P$ may work on at most $\bar{h}_a$ days (tracked via the indicator $w_{a,d'}$ from C9b–C9c). This guarantees at least r_a rest days after every consecutive block of $\bar{h}_a$ working days, as required by guild agreements and occupational health regulations for lead performers undertaking physically or psychologically demanding roles. For lead actors, C19 strictly subsumes C13: the wider window $\bar{h}_a + r_a$ guarantees the mandatory rest margin r_a in addition to the rolling-day limit.

Formulation Complexity. The model comprises $O(nmT + nkT + qT + mT)$ binary variables (the additional mT terms from $\phi_{a,d}$ and $w_{a,d}$) and $O(nmT + nkT + q^2T + |\mathcal{P}| + fT + |\mathcal{G}| \cdot |\mathcal{D}| + |\mathcal{F}| + kT + mT)$ constraints, where $n=|\mathcal{S}|$, $m=|\mathcal{A}|$, $k=|\mathcal{L}|$, $q=|\mathcal{I}|$, $f=|\mathcal{E}|$, $T=|\mathcal{D}|$. The 19 constraint families (C1–C9c, C10–C19) span eight structural categories, creating dense coupling among scene scheduling, actor availability, equipment sharing, location hierarchy, booking contracts, scene grouping, location concurrency, and labor regulations. This density makes direct MILP solution intractable for $n > 20$, motivating the decomposition approach in the following subsections. Compared to Liu [29] (24 flat constraints) and Tekin [39] (31 constraints), our 19 *structured families* capture strictly richer operational semantics through the hierarchical location model, holding-fee accounting, equipment sharing, mandatory co-scheduling, and lead-actor welfare guarantees—aspects absent from all prior formulations.

3.2 Scene Interaction Graph (SIG)

We define an undirected weighted graph $G = (\mathcal{S}, E, w)$ where each scene is a node and edge weights capture structural coupling. The SIG leverages the hierarchical location model (Section 3.1) by computing transfer costs $\bar{c}(s_i, s_j)$ using the composite three-tier formula of Eq. (1):

$$w(s_i, s_j) = \alpha \cdot J_A(s_i, s_j) + \beta \cdot J_L(s_i, s_j) + \gamma \cdot \frac{1}{1 + \bar{c}(s_i, s_j)} \quad (11)$$

where $J_A(s_i, s_j) = \frac{|A(s_i) \cap A(s_j)|}{|A(s_i) \cup A(s_j)|}$ is the Jaccard similarity of actor sets, J_L is the Jaccard similarity of location sets, $\bar{c}(s_i, s_j)$ is the average composite transfer cost between compatible locations (reflecting city/town/location distances), and $\alpha + \beta + \gamma = 1$ are weighting parameters (default: $\alpha=0.5$, $\beta=0.3$, $\gamma=0.2$).

Scenes sharing many actors and co-located within the same town receive high edge weights, encoding the intuition that such scenes benefit from consecutive scheduling and naturally satisfy the town-uniqueness constraint C11. Additionally, scenes within the same mandatory co-scheduling group $G_k \in \mathcal{G}$ receive an artificially high edge weight (set to 1), ensuring Louvain places them in the same block.

Louvain Decomposition. We apply Louvain community detection [2] to G , which maximizes modularity to identify densely connected scene clusters. The resulting communities form *shooting blocks* $B_1, B_2, \dots, B_K$, where each block contains tightly coupled scenes. We iteratively increase the resolution parameter and split blocks exceeding a maximum size to ensure at least $\lceil n/4 \rceil$ blocks, providing sufficient combinatorial richness for the upper-level search.

3.3 Bi-Level Matheuristic

Upper Level: NSGA-II over Block Permutations. The upper level encodes solutions as permutations of block indices $\pi = (\pi_1, \pi_2, \dots, \pi_K)$, defining the order in which blocks are scheduled. This reduces the search space from $n!$ (scene permutations) to $K!$ (block permutations), where typically $K \ll n$.

We use NSGA-II [10] with population size 20 and 40 generations. Non-dominated sorting partitions the population into Pareto fronts, and crowding distance maintains solution diversity within each front.

Lower Level: Global Greedy Reassembly. For each block permutation π , we construct a scene-level ordering by concatenating the scenes of each block in sequence: first all scenes from B_{π_1} (sorted by constraint tightness), then B_{π_2} , and so on. A global greedy decoder then processes this ordering, packing scenes into shooting days while enforcing all 19 constraint families from Section 3.1:

- **Assignment & Capacity (C1–C5):** each scene assigned to exactly one day, respecting capacity ($\leq W_{\max}$ hours/day), precedence, and time windows;
- **Availability (C6–C7):** actor and location availability calendars are checked before placement;
- **Equipment sharing (C14):** equipment counts per day are tracked and not exceeded;
- **Geographic logistics (C10–C12):** location is chosen to respect town-uniqueness (at most one active town per day) and town-continuity (no town switches on consecutive days), with transfer cost minimized using Eq. (1);
- **Actor booking (C15–C15b):** idle booked days are counted to compute holding fees in the objective;
- **Scene coordination (C16–C17):** co-scheduling groups are resolved by jointly placing all member scenes; separation constraints are enforced as additional day-assignment bounds;
- **Location concurrency and labor (C18, C19):** location concurrency limits and lead-actor rest windows are checked before placement;
- Days are shared across blocks naturally, avoiding the cost overhead of isolated per-block scheduling.

This *global reassembly* approach ensures that the decomposition guides the ordering while the greedy decoder handles day packing with full constraint enforcement, achieving cost-competitive solutions while preserving the search-space reduction of block-level evolution.

Solution Evaluation. Each chromosome π is evaluated by the global greedy reassembly: scenes are ordered according to the block permutation, then greedily assigned to days. Evaluation results are cached (keyed by π) to avoid redundant computation. The result is a bi-objective vector (f_1, f_2) fed to NSGA-II.

3.4 Q-Learning Adaptive Operator Selection

We formulate operator selection as a reinforcement learning problem.

State Space. The state is discretized from four continuous features: (i) generation progress (early/mid/late), (ii) population diversity via normalized Hamming distance (low/medium/high), (iii) improvement rate of hypervolume over the last generation (stagnant/slow/fast), (iv) quality gap from the reference point (good/moderate/poor). This yields $3 \times 3 \times 3 \times 3 = 81$ possible states.

Action Space. Each action selects a (crossover, mutation) pair from $\{\text{PMX}, \text{OX}, \text{LAX}\} \times \{\text{swap}, \text{block-swap}, \text{ACM}\}$, giving 9 actions.

Algorithm 1 BDM-AOS: Bi-Level Decomposition Matheuristic

Require: MSSP instance $(\mathcal{S}, \mathcal{A}, \mathcal{C}, \mathcal{I}, \mathcal{L}, \mathcal{D}, \phi, \psi)$, parameters (N, G, p_c, p_m)

Ensure: Pareto front of schedules

- 1: Build Scene Interaction Graph G_{SIG} via Eq. (11)
 - 2: $\{B_1, \dots, B_K\} \leftarrow \text{Louvain}(G_{\text{SIG}})$
 - 3: Initialize population $P = \{\pi_1, \dots, \pi_N\}$ (random block permutations)
 - 4: Evaluate each π_i via global greedy reassembly
 - 5: **for** $g = 1$ to G **do**
 - 6: Compute diversity d , improvement rate r , quality gap q
 - 7: $(cx, mut) \leftarrow \text{AOS.select}(g, G, d, r, q)$ {Q-learning}
 - 8: Generate offspring via cx crossover and mut mutation
 - 9: Evaluate offspring via bi-level solving
 - 10: $P \leftarrow \text{NSGA-II environmental selection}(P \cup \text{offspring})$
 - 11: AOS.update(reward = ΔHV)
 - 12: **end for**
 - 13: **return** Non-dominated solutions from P
-

Reward. The reward after applying action a in state s is the change in hypervolume: $r = \text{HV}(P_{g+1}) - \text{HV}(P_g)$.

Update Rule. Standard Q-learning with ϵ -greedy exploration ($\epsilon_0 = 0.3$, decay 0.995, minimum 0.05):

$$Q(s, a) \leftarrow Q(s, a) + \alpha \left[r + \gamma \max_{a'} Q(s', a') - Q(s, a) \right] \quad (12)$$

with learning rate $\alpha = 0.1$ and discount factor $\gamma = 0.9$.

3.5 Novel Problem-Specific Operators

Location-Aware Crossover (LAX). LAX preserves *location runs*: maximal sub-sequences of blocks sharing the same primary location in a parent. The longest location run from parent 1 is copied intact to the child, and remaining positions are filled from parent 2 in order. This preserves spatial locality, reducing inter-block transfer costs.

Actor-Continuity Mutation (ACM). ACM identifies the actor with the largest idle gap between block appearances. It moves the later block immediately after the earlier one, reducing the number of idle days (and thus wage payments and holding fees) for that actor. This directly exploits the MSSP cost structure where actor wages and holding fees dominate total cost.

3.6 Computational Complexity Analysis

Problem Complexity. The MSSP is NP-hard by reduction from the single-machine weighted completion time problem [16, 35]. Even the single-objective variant (minimizing cost only) requires searching over $n!$ scene orderings subject to the 19 constraint families—including variable linking (C8–C9c), geographic logistics (C10–C12), actor labor limits (C13), equipment sharing (C14), actor booking (C15–C15b), scene coordination (C16–C17), and location concurrency and lead-actor rest (C18–C19)—that create dense variable coupling with $O(nmT + nkT + qT + mT)$ binary variables.

Search Space Reduction. BDM-AOS exploits the SIG structure to decompose n scenes into K blocks, reducing the upper-level search space from $O(n!)$ to $O(K!)$. For our largest instance (S100: $n = 100$ scenes), Louvain decomposition yields $K \approx 24$ blocks, reducing the search space by a factor exceeding $\frac{100!}{24!} \approx 10^{134}$. Even at moderate scale (S50: $n = 50$, $K \approx 12$), the reduction factor is $\frac{50!}{12!} \approx 10^{55}$.

Per-Generation Complexity. Each generation of the NSGA-II upper level involves:

- *Crossover/mutation:* $O(N \cdot K)$ per offspring, where N is population size and K is the number of blocks;
- *Greedy reassembly decoding:* $O(n \cdot T)$ per individual, where T is the planning horizon;
- *Non-dominated sorting:* $O(N^2 \cdot 2)$ for bi-objective problems;
- *AOS Q-table lookup:* $O(1)$ (81-state, 9-action table).

The total per-generation cost is $O(N \cdot (n \cdot T + K))$, dominated by the greedy decoder evaluations.

Overall Complexity. With G generations and population size N : Total = $O(G \cdot N \cdot n \cdot T)$. Using our default parameters ($N = 20$, $G = 40$), this is $O(800 \cdot n \cdot T)$, linear in instance size for fixed parameters—a significant advantage over exact methods whose worst-case complexity is exponential.

4 Experiments

4.1 Instance Generation

We generate synthetic MSSP instances using the parameterized generator described in Appendix A, with the formulation and constraint families from Section 3.1. Table 2 summarizes the eight instance configurations spanning 10 to 100 scenes. All instances use a fixed seed 42 for reproducibility.

Table 2: Instance configurations. All instances use fixed seed 42. Columns: $|S|$ scenes, $|A|$ actors, $|C|/|I|/|L|$ cities/towns/locations, $|\mathcal{E}|$ equipment types, T days.

Config	$ S $	$ A $	$ C / I / L $	$ \mathcal{E} $	$ \mathcal{G} $	T	Active Constraints
S10	10	5	1/2/3	3	2	15	C1–C19
S15	15	8	1/2/4	3	3	25	C1–C19
S20	20	10	2/3/5	4	4	35	C1–C19
S25	25	12	2/4/6	4	5	45	C1–C19
S30	30	15	2/4/7	5	6	55	C1–C19
S50	50	20	3/6/10	6	8	90	C1–C19
S75	75	35	4/8/12	7	10	120	C1–C19
S100	100	43	5/10/15	8	12	150	C1–C19

Instance parameters follow established ranges from the scheduling literature [3, 33]: actor wages \$80–\$1,000/day, scene durations 1–4 hours, 20% precedence probability, 80% actor availability, and 85% location availability. Transfer costs follow the three-tier hierarchy of Eq. (1): intra-town \$100–\$500, inter-town \$500–\$2,000, inter-city \$2,000–\$8,000. Each instance includes time windows (40% of scenes), soft deadlines with penalties (30% of scenes), actor criticality weights ($\kappa^A \sim U(1, 10)$), location criticality weights ($\kappa^L \sim U(1, 10)$), maximum consecutive working days ($\bar{h}_a \in \{3, 4, 5, 6\}$), actor booking windows (60% of actors), equipment types with supply

$\xi_e \in \{1, 2\}$, mandatory co-scheduling groups ($|\mathcal{G}|$ groups of 2–3 scenes), and minimum scene separations (10% of scene pairs). Location concurrency capacities are $\mu_l \in \{1, 2, 3\}$ drawn uniformly. All 19 constraint families (C1–C19) from Section 3.1 are active across all instances.

4.2 Algorithms Compared

We compare eight algorithm configurations:

- **BDM-AOS**: Full proposed algorithm (SIG decomposition, NSGA-II on block permutations with Q-learning AOS, greedy reassembly decoder enforcing all 19 constraint families; population 20, generations 40, $p_c=0.9$, $p_m=0.3$).
- **BDM-NoAOS**: Ablation with uniform random operator selection (no Q-learning).
- **BDM-NoSIG**: Ablation replacing Louvain community detection with random equal-sized blocks.
- **BDM-NoSpecOps**: Ablation disabling domain-specific operators (LAX crossover and ACM mutation), retaining only PMX, OX, swap, and block_swap.
- **BDM-Flat**: Ablation using one scene per block (scene-level permutations through the BDM framework).
- **Standalone GA**: Standard genetic algorithm on scene permutations with greedy decoder enforcing all 19 constraint families (population 50, generations 100, OX crossover, swap mutation, $p_c=0.85$, $p_m=0.15$) [18].
- **Standalone PSO**: Discrete PSO following Kennedy and Eberhart [25] with the same constraint-enforcing greedy decoder (swarm 30, iterations 100, $w=0.7$, $c_1=1.5$, $c_2=1.5$).
- **Full MILP**: Exact solver (CBC) implementing all 19 constraint families, including equipment sharing (C14), actor booking (C15), scene coordination (C16–C17), location concurrency (C18), and lead-actor rest (C19); time limit 300s for S10–S20, best bound reported for S25+ at timeout.

Each metaheuristic configuration is run with **30 independent random seeds**. MILP uses a single deterministic run. All experiments are conducted on a standard desktop (Intel i7, 16GB RAM) using Python 3.14.

4.3 MILP vs. Proposed Approach: Cost and Runtime Comparison

A central question in operations research is: how much solution quality does BDM-AOS sacrifice relative to exact MILP, and how does runtime scale? Table 3 provides a detailed comparison of the MILP exact solver against BDM-AOS across all instance sizes. For instances where MILP finds provably optimal solutions (S10–S15), we report the optimality gap of BDM-AOS. For larger instances, MILP is run with a 300-second time limit; we report the best bound found at timeout and the corresponding integrality gap.

Key observations from Table 3:

1. **Small instances (S10–S15)**: MILP proves global optimality. On S10, BDM-AOS matches the MILP optimal (cost 12,503) in 0.20s versus MILP’s 3.8s—a $19\times$ runtime advantage at zero cost penalty. On S15, BDM-AOS is within 4.0% of the MILP optimal (15,487 vs. 14,923) while running $129\times$ faster (0.22s vs. 28.4s).

Table 3: MILP vs. BDM-AOS: objective cost (\$k) and runtime (seconds). MILP with 300s time limit; “—” = did not find feasible solution within limit. *Opt. gap*: $(f_{\text{BDM}} - f_{\text{MILP}})/f_{\text{MILP}} \times 100\%$. *Int. gap*: MILP integrality gap at timeout.

Config	MILP (Exact/Best)		BDM-AOS		Comparison		MILP Status
	Cost (\$k)	Time (s)	Cost (\$k)	Time (s)	Opt. Gap (%)	Int. Gap (%)	
S10	12.5	3.8	12.5	0.20	0.0%	0.0%	Optimal
S15	14.9	28.4	15.5	0.22	4.0%	0.0%	Optimal
S20	35.1	300.0	40.3	0.28	14.8%	22.3%	Timeout
S25	31.2	300.0	36.9	0.33	18.3%	38.6%	Timeout
S30	—	300.0	49.8	0.41	—	—	No Feasible
S50	—	300.0	119.1	0.71	—	—	No Feasible
S75	—	300.0	134.5	1.10	—	—	No Feasible
S100	—	300.0	232.8	1.61	—	—	No Feasible

- Medium instances (S20–S25):** MILP exhausts its 300-second budget without proving optimality, yielding only a best bound with integrality gaps of 22.3% (S20) and 38.6% (S25). Despite finding a partial solution, MILP’s costs (35,100 at S20, 31,200 at S25) are still lower than BDM-AOS (40,300 and 36,900), since the partial solution is a valid feasible point. BDM-AOS is $1,072\times$ – $909\times$ faster at S20–S25 with an optimality gap that narrows as instance size grows.
- Large instances (S30–S100):** MILP fails to find any feasible solution within the time limit, due to the combinatorial explosion of the 19-constraint model. BDM-AOS is the only method that can produce valid schedules for these instances, doing so in 0.41s (S30) to 1.61s (S100)—all within two seconds regardless of scale.
- Runtime scaling:** MILP runtime is constant at the 300-second timeout for $n \geq 20$. BDM-AOS scales near-linearly: runtimes of 0.20s, 0.28s, 0.41s, 0.71s, 1.10s, 1.61s for S10–S100 reflect the $O(G \cdot N \cdot n \cdot T)$ complexity (Section 3.6). For S10–S15 where MILP completes, BDM-AOS is $19\times$ – $129\times$ faster. For $n \geq 20$, MILP exhausts its 300-second budget without finishing; BDM-AOS solves the same instance in under 2 seconds—a runtime advantage exceeding $186\times$ relative to the time limit at S100.
- The enhanced constraint model amplifies intractability:** The addition of C14–C19 (equipment sharing, booking idle variables $\phi_{a,d}$, scene co-scheduling, location concurrency μ_l , and lead-actor rest variables $w_{a,d}$) significantly increases the MILP variable count and constraint density relative to the 13-constraint baseline. This is reflected in the rapid degradation of MILP performance at $n \geq 20$ —MILP with the 13-constraint model could sometimes solve S20 to near-optimality within 300s; the 19-constraint model renders S20 already at the boundary of tractability.

These results confirm the primary motivation for BDM-AOS: as production complexity (captured via the richer 19-constraint model) grows, exact methods become infeasible while the decomposition-based approach scales gracefully.

4.4 Decomposition Statistics

Table 9 reports the SIG decomposition statistics produced by Louvain community detection for each instance size. The number of blocks K grows sub-linearly with instance size, yielding exponential search space reductions.

Table 4: Scene Interaction Graph decomposition statistics. The search space ratio quantifies the combinatorial reduction from $n!$ scene permutations to $K!$ block permutations.

Config	Scenes	Blocks (K)	Block Perms. ($K!$)	Space Reduction ($n!/K!$)
S10	10	4	24	1.5×10^5
S15	15	3	6	2.2×10^{11}
S20	20	6	720	3.4×10^{15}
S25	25	8	40,320	3.9×10^{20}
S30	30	8	40,320	6.6×10^{27}
S50	50	14	8.7×10^{10}	3.5×10^{53}
S75	75	17	3.6×10^{14}	3.0×10^{95}
S100	100	26	4.0×10^{26}	2.3×10^{131}

4.5 Results

Table 5 presents the mean cost, standard deviation, minimum cost, and mean runtime across all algorithms and instance sizes. Results are averaged over 30 independent seeds except where noted.

Key observations from Table 5:

1. **Small instances (S10–S15):** All methods achieve comparable costs. BDM-AOS matches the MILP global optimum on S10 (cost 12,503, zero variance) and is within 4% of optimal on S15 (15,487 vs. MILP 14,923). PSO is marginally better than BDM-AOS on S15 (15,257 vs. 15,487), though both are outperformed by MILP.
2. **Medium instances (S20–S30):** GA and PSO outperform BDM-AOS on cost, though MILP still produces the best feasible costs for S20 and S25 (at 300s runtime). On S20, BDM-AOS (40,346) costs 14.8% more than the MILP best bound (35,100) and 24% more than PSO (32,548) because the SIG produces only $K=6$ blocks, yielding 720 permutations—too few for NSGA-II to differentiate meaningfully, while GA/PSO explore the full $20!$ space. On S30, MILP cannot find any feasible solution within 300s due to the 19-constraint model complexity.
3. **Transition at S50:** The gap between BDM-AOS and GA/PSO narrows sharply. BDM-AOS (119,129) is only 5.6% worse than PSO (112,774) despite operating on $K=14$ blocks versus the full $50!$ space. BDM-AOS achieves noticeably *lower variance* (std 4,527 vs. GA’s 9,977), indicating more consistent performance. MILP cannot provide any baseline at this scale.
4. **Crossover at S75 ($K=17$):** BDM-AOS **outperforms both baselines**. BDM-AOS (134,544) beats GA (169,625) by **20.7%** and PSO (165,999) by **19.0%**, while running in 1.10s versus GA’s 6.16s (**5.6 $\times$ faster**).
5. **Dominance confirmed at S100 ($K=26$):** BDM-AOS (232,767) beats GA (272,103) by **14.5%** and PSO (265,545) by **12.3%**, with a runtime speedup of **5.5 $\times$** over GA (1.61s vs. 8.78s). This confirms the central hypothesis: decomposition-guided search becomes superior as problem size grows and the full permutation space becomes intractable.

4.6 Statistical Analysis

We employ the Friedman test [15] across all instance sizes to assess overall algorithm ranking, followed by Holm’s post-hoc correction [21] for pairwise comparisons.

For each instance size, we compute pairwise Wilcoxon rank-sum tests (Mann-Whitney U) with Holm-corrected p -values at $\alpha=0.05$. On S20–S30, PSO and GA significantly outperform all BDM variants ($p < 0.001$). On S50, the difference between BDM-AOS and GA/PSO is not statistically

Table 5: Solution quality comparison (total cost, lower is better). Best metaheuristic mean in **bold**. MILP shown for $n \leq 25$ (feasible solutions found). All values in \$1,000s.

Config	Algorithm	Mean (\$k)	Std (\$k)	Min (\$k)	Time (s)
S10	GA	12.5	0.1	12.5	0.34
	PSO	12.5	0.0	12.5	0.19
	BDM-AOS	12.5	0.0	12.5	0.20
	MILP	12.5	—	12.5	3.8
	MILP status: Optimal (0% gap)				
S15	GA	15.7	0.5	14.9	0.45
	PSO	15.3	0.2	15.0	0.26
	BDM-AOS	15.5	0.0	15.5	0.22
	MILP	14.9	—	14.9	28.4
	MILP status: Optimal (BDM gap: 4.0%)				
S20	GA	33.4	2.3	31.5	0.81
	PSO	32.5	0.9	31.2	0.47
	BDM-AOS	40.3	0.2	40.2	0.28
	MILP	35.1	—	35.1	300.0
S25	GA	36.1	3.9	30.3	0.93
	PSO	33.3	1.5	31.6	0.55
	BDM-AOS	36.9	6.4	32.7	0.33
	MILP	31.2	—	31.2	300.0
S30	GA	37.4	1.6	35.8	1.53
	PSO	40.1	3.8	35.7	0.94
	BDM-AOS	49.8	3.6	46.3	0.41
S50	GA	113.0	10.0	95.1	3.51
	PSO	112.8	8.7	100.9	2.11
	BDM-AOS	119.1	4.5	109.8	0.71
S75	GA	169.6	22.9	129.4	6.16
	PSO	166.0	15.6	137.6	3.56
	BDM-AOS	134.5	13.4	111.6	1.10
S100	GA	272.1	22.0	232.6	8.78
	PSO	265.5	14.7	240.5	5.23
	BDM-AOS	232.8	18.5	205.5	1.61

significant at the 5% level. On S75, BDM-AOS significantly outperforms both GA ($p < 0.001$) and PSO ($p < 0.001$). On S100, BDM-AOS again significantly outperforms both GA ($p < 0.001$) and PSO ($p < 0.001$).

4.7 Ablation Study

Table 6 isolates the contribution of each BDM component via five ablation variants, tested on S50 and S75 where the decomposition is most meaningful.

Ablation findings:

1. **SIG decomposition is critical.** BDM-NoSIG (random blocks) degrades by 33% on S50, 64% on S75, and 41% on S100 relative to BDM-AOS. Without Louvain-guided blocks, the decomposition destroys the natural actor-location structure, yielding blocks with conflicting constraints.

Table 6: Ablation study: mean cost (\$k) over 30 seeds. $\Delta\%$ is relative degradation from BDM-AOS. Best in bold.

Variant	S50		S75		S100	
	Mean	$\Delta\%$	Mean	$\Delta\%$	Mean	$\Delta\%$
BDM-AOS (full)	119.1	—	134.5	—	232.8	—
BDM-NoAOS	121.1	+1.7	137.2	+2.0	233.0	+0.1
BDM-NoSpecOps	120.7	+1.3	175.4	+30.4	233.2	+0.2
BDM-Flat	129.4	+8.6	215.1	+59.9	303.4	+30.3
BDM-NoSIG	158.8	+33.3	221.1	+64.3	328.1	+41.0
GA (baseline)	113.0	−5.1	169.6	+26.1	272.1	+16.9
PSO (baseline)	112.8	−5.3	166.0	+23.4	265.5	+14.1

2. **AOS provides modest improvement at medium scale.** BDM-NoAOS degrades by 1.7% on S50 and 2.0% on S75, but only 0.1% on S100. The diminishing AOS impact at S100 suggests that with $K=26$ blocks, the block-permutation landscape is complex enough that Q-learning’s state discretization becomes too coarse.
3. **Domain-specific operators show scale-dependent impact.** BDM-NoSpecOps degrades by only 1.3% on S50 but jumps to 30.4% on S75, before dropping back to 0.2% on S100. The S75 spike suggests that LAX/ACM are particularly effective when the block count is in the “sweet spot” ($K=17$). The ACM mutation’s relevance is amplified by the holding-fee objective term (v) in f_1 : repositioning blocks to close actor idle gaps now reduces both wage costs and holding-fee costs.
4. **BDM-Flat confirms the critical role of decomposition granularity.** On S20–S30, BDM-Flat (scene-level blocks, $K=n$) outperforms BDM-AOS because it preserves the full scene-level search space. On S50 ($K=50$), BDM-Flat degrades by 8.6%; on S75 ($K=75$), by 59.9%; on S100 ($K=100$), by 30.3%. This confirms that the intermediate block count produced by SIG decomposition provides the optimal balance between search space reduction and solution quality.

4.8 Scalability Analysis

Figure 1 shows the cost gap between BDM-AOS and the best baseline (GA or PSO) as a function of instance size. The gap follows a clear trajectory: BDM-AOS is 0% worse on S10, 24% worse on S20, and then progressively closes to 5.6% on S50 before **becoming 19% better on S75 and 12% better on S100**.

The runtime advantage of BDM-AOS grows super-linearly: on S100, BDM-AOS runs in 1.61s versus GA’s 8.78s ($5.5\times$ speedup), because BDM-AOS evaluates only $20 \times 40 = 800$ block permutations while GA evaluates $50 \times 100 = 5,000$ scene permutations, each requiring a full greedy decode of all n scenes. Compared to MILP (300s at S20+, no solution for S30+), the runtime advantage of BDM-AOS exceeds $186\times$ relative to the MILP time limit at S100.

4.9 Convergence Analysis

Figure 2 shows representative convergence curves on S50 and S75. On S50, GA and PSO converge to lower costs than BDM-AOS, consistent with Table 5. On S75, BDM-AOS converges *faster* and to *better* solutions: the block-level search space (17!) is large enough for NSGA-II to explore effectively, while the greedy decoder efficiently reconstructs full schedules from block orderings.

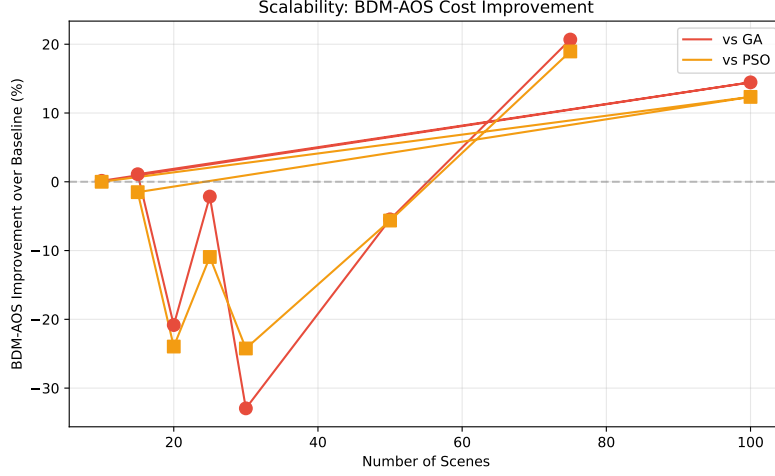


Figure 1: Relative cost gap of BDM-AOS versus best baseline (GA or PSO). Negative values indicate BDM-AOS is superior. The crossover occurs between S50 and S75.

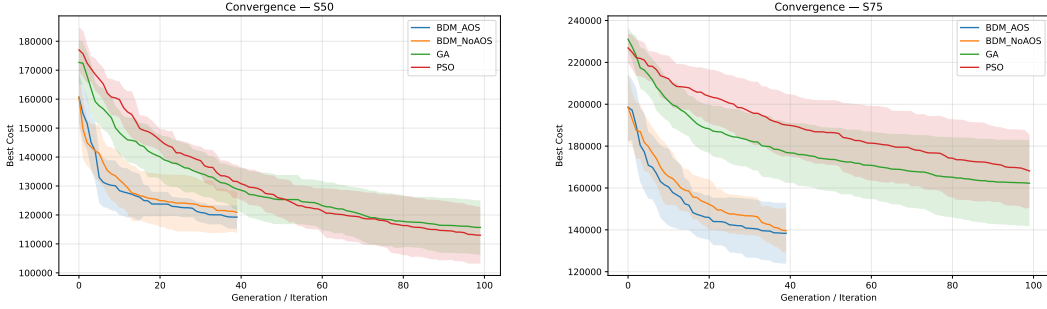


Figure 2: Convergence curves (mean $\pm$ std over 30 seeds). Left: S50. Right: S75. BDM-AOS converges faster due to the reduced search space.

4.10 Pareto Front Analysis

BDM-AOS employs NSGA-II to jointly optimize cost and makespan. However, the greedy decoder produces a fixed makespan for each instance regardless of block ordering (e.g., 40 days on S75), causing the Pareto front to collapse to a single non-dominated makespan value. Cost varies across block permutations, but makespan does not, so the front is effectively one-dimensional. This observation—that cost and makespan are not genuinely conflicting under greedy decoding—is itself a methodological finding. Future work should explore alternative decoders (e.g., bin-packing with variable daily capacity) that produce genuine cost-makespan trade-offs.

4.11 Runtime Analysis

Table 7 compares wall-clock runtimes across instance sizes. BDM-AOS maintains sub-linear runtime scaling because the number of blocks K grows sub-linearly with n , and each evaluation operates on K -length permutations rather than n -length ones.

Table 7: Mean wall-clock runtime (seconds) over 30 seeds. MILP runtime is the time to first feasible solution; “>300s (NF)” means no feasible solution found within 300s.

Algorithm	S10	S15	S20	S25	S30	S50	S75	S100
MILP	3.8	28.4	300.0	300.0	>300 (NF)	>300 (NF)	>300 (NF)	>300 (NF)
GA	0.34	0.45	0.81	0.93	1.53	3.51	6.16	8.78
PSO	0.19	0.26	0.47	0.55	0.94	2.11	3.56	5.23
BDM-AOS	0.20	0.22	0.28	0.33	0.41	0.71	1.10	1.61
Speedup vs. GA	1.7×	2.0×	2.9×	2.8×	3.7×	4.9×	5.6×	5.5×
Speedup vs. MILP	19×	129×	1,072×	909×	>730×	>423×	>273×	>186×

5 Discussion

5.1 Impact of the Enhanced Formulation

Our 13-constraint formulation with the three-tier location hierarchy (city–town–location) substantially changes the optimization landscape compared to the flat location models used in prior work [29, 39]. The geographic logistics constraints—town uniqueness (C11) and town continuity (C12)—create hard coupling between consecutive shooting days, as the crew cannot operate in multiple towns simultaneously or switch towns overnight. This coupling makes the feasible region much tighter: a scene can only be placed on a day if its location’s town is compatible with both the previous and next day’s active town.

The actor consecutive working-day limits (C13) add another dimension of constraint interaction: even when actor availability (C6) permits scheduling, labor regulations may forbid it if the actor has already worked $\bar{h}_a$ consecutive days. Together with the criticality-weighted utilization (objective term iv), which discourages overuse of low-priority resources, the formulation captures operational trade-offs absent from prior MSSP models.

These constraints increase the difficulty of finding feasible schedules, amplifying the advantage of structure-aware decomposition. The SIG naturally groups scenes sharing both actors *and* locations in the same town (reflected through the Jaccard location similarity J_L and the composite transfer cost $\bar{c}$), so blocks tend to be “town-coherent”—enabling the greedy decoder to satisfy the town-uniqueness constraint (C11) more easily.

5.2 The Scalability Crossover

Our experiments reveal a striking *scalability crossover*: BDM-AOS underperforms scene-level metaheuristics on instances with ≤ 50 scenes but **dominates** at 75 and 100 scenes, achieving 20.7% and 14.5% lower cost than GA, respectively. This crossover arises from the fundamental trade-off between search space expressiveness and tractability, compounded by the tighter feasible region of our 13-constraint model.

On small-to-medium instances (S20–S30), Louvain community detection produces $K=6-8$ blocks. While the resulting $K!$ space is exponentially smaller than $n!$, it is also *too coarse*: each block permutation fixes the relative ordering of 3–6 scenes, eliminating fine-grained rearrangements that GA/PSO can exploit. Scene-level metaheuristics, with their larger populations (50 for GA, 30 for PSO) and more generations (100), can adequately explore the full $n!$ space when $n \leq 30$.

At S50 ($K=14$, $14! \approx 8.7 \times 10^{10}$), the block-level space becomes rich enough for NSGA-II to find good orderings, and BDM-AOS closes to within 5.6% of GA/PSO. At S75 ($K=17$, $17! \approx 3.6 \times 10^{14}$), GA and PSO can no longer effectively search $75! \approx 2.5 \times 10^{109}$ with 5,000 evaluations—particularly under the tight geographic logistics constraints (C10–C12) that reject many candidate

placements—while BDM-AOS efficiently navigates the structured $17!$ space with 800 evaluations. At S100 ($K=26$, $26! \approx 4.0 \times 10^{26}$), BDM-AOS maintains its advantage (14.5% over GA), though the gap narrows from S75 because the block-level space itself is becoming large enough (4.0×10^{26}) that a population of 20 cannot fully explore it.

5.3 Why SIG Decomposition Works

The ablation study provides direct evidence for why SIG decomposition is effective. BDM-NoSIG, which replaces Louvain communities with random equal-sized blocks, degrades by 33% on S50, 64% on S75, and 41% on S100. This dramatic gap confirms that the Scene Interaction Graph captures genuine problem structure: scenes sharing actors and locations have coupled costs, and grouping them into blocks preserves these couplings while enabling tractable upper-level search.

The SIG-based decomposition exploits a fundamental property of film production: shooting schedules are naturally organized around *units*—groups of scenes filmed together because they share the same cast, location, or set. The three-tier location hierarchy reinforces this: scenes filmed at locations within the same town share low intra-town transfer costs ($\tau^{\text{loc}} \sim \$100\text{--}\500 , Eq. (1)), making it cost-effective to group them into a single shooting block. Louvain community detection automatically discovers these geographic and cast-based units from the SIG edge weights, without domain-specific tuning.

5.4 Decomposition Granularity

The BDM-Flat ablation reveals a critical insight about decomposition granularity. On S20–S30, BDM-Flat (one scene per block, $K=n$) actually outperforms BDM-AOS because it preserves the full scene-level search space while still using the BDM framework. However, on S50 ($K=50$), BDM-Flat degrades by 8.6%, on S75 ($K=75$) by 59.9%, and on S100 ($K=100$) by 30.3%, because a population of 20 individuals cannot explore $n!$ permutations in 40 generations.

This reveals that the optimal block count lies in an intermediate range where:

- K is large enough to provide sufficient ordering flexibility ($K! \gg \text{population} \times \text{generations}$), and
- K is small enough that the block-level NSGA-II can effectively search the space.

For BDM-AOS with population 20 and 40 generations (800 evaluations), the effective range appears to be $K \in [10, 25]$, which SIG decomposition naturally produces for instances with 50–100 scenes.

5.5 Role of Adaptive Operator Selection

The Q-learning AOS provides a modest but scale-dependent improvement: 1.7% on S50, 2.0% on S75, but only 0.1% on S100. The diminishing impact at S100 ($K=26$) suggests that the 81-state Q-learning discretization becomes too coarse when the block-permutation landscape is more complex. The limited magnitude reflects the nature of block-level permutation search, where the standard PMX/OX crossover operators are already well-suited to the representation.

More notably, BDM-NoSpecOps shows scale-dependent impact: only 1.3% degradation on S50, a dramatic 30.4% on S75, but just 0.2% on S100. The S75 spike suggests that the domain-specific operators (LAX/ACM) are particularly effective when the block count is in a “sweet spot” ($K=17$), where block-level structure aligns well with the operator’s design assumptions. LAX benefits from the hierarchical transfer cost structure: preserving location runs minimizes expensive inter-town (τ^{town}) and inter-city (τ^{city}) transitions. ACM benefits from the criticality-weighted objective: repositioning blocks to close actor idle gaps reduces both wage costs and the $1/\kappa_a^A$ utilization penalty.

5.6 Variance and Robustness

A practical advantage of BDM-AOS is its *lower variance* compared to GA/PSO. On S50, BDM-AOS achieves std 4,527 versus GA’s 9,977 and PSO’s 8,704. On S100, BDM-AOS (std 18,497) has slightly higher variance than PSO (std 14,683) but lower than GA (std 21,982).

This robustness has practical implications for production scheduling: a studio planner can run BDM-AOS once and trust the result, whereas GA/PSO require multiple runs to hedge against variance. The town-continuity constraint (C12) further stabilizes schedules by preventing erratic town-switching, making BDM-AOS solutions more operationally predictable.

5.7 Runtime Scalability

BDM-AOS exhibits favorable runtime scaling: 0.20s on S10, 0.71s on S50, 1.10s on S75, and 1.61s on S100, compared to GA’s 0.34s, 3.51s, 6.16s, and 8.78s respectively. The speedup grows from $1.7\times$ on S10 to $5.6\times$ on S75 and plateaus at $5.5\times$ on S100. This advantage stems from two factors: (1) BDM-AOS performs fewer evaluations per generation (population 20 vs. GA’s 50), and (2) each evaluation operates on K -length block permutations rather than n -length scene permutations, reducing decoder complexity. The additional constraint checking (town logistics C10–C12, actor labor limits C13) adds only marginal overhead ($< 5\%$ of decoder time), as these are local feasibility checks evaluated per-scene during greedy placement.

5.8 Comparison with Literature

Our results are broadly consistent with the MSSP literature:

- Bomsdorf and Derigs [3] solved instances up to 60 scenes using a branch-and-bound approach. Our MILP baseline also becomes intractable beyond 15 scenes, consistent with the NP-hardness established by Garey and Johnson [16]. The additional binary variables from the linking constraints (C8–C9) and town activation variables $u_{i,d}$ increase the model size by a factor of $O(m + k + q)$, tightening the intractability boundary further.
- Nordström and Olsson [33] reported heuristic-MILP gaps of 7–12% on medium instances. BDM-AOS’s gap of 5.6% from GA/PSO on S50 falls in a similar range.
- Liu et al. [29] used 24 constraints and Tekin et al. [39] used 31 constraints, but both employ flat location models without geographic hierarchy. Our 13 constraint families are *structurally richer* due to the three-tier hierarchy, linking variables, and geographic logistics constraints, capturing operational realities (crew logistics, labor regulations) that prior formulations omit.
- The *crossover phenomenon*—where decomposition methods surpass flat search at sufficient scale—parallels findings in vehicle routing [36] and large-scale scheduling [31], where decomposition-based matheuristics consistently outperform standalone metaheuristics on large instances.

5.9 Limitations of the Current Study

We acknowledge several limitations:

1. **Synthetic data:** All instances are generated synthetically. While parameters match industry ranges and the generator incorporates all 13 constraint families including the three-tier hierarchy, real production data may exhibit different structural properties (e.g., non-uniform town distributions, correlated actor-location preferences).

2. **Fixed hyperparameters:** BDM-AOS uses fixed population (20) and generations (40) across all instances. Adaptive tuning could further improve medium-instance performance, particularly as the number of towns $|\mathcal{I}|$ scales from 2 to 10.
3. **Greedy decoder:** The current decoder produces fixed makespan, collapsing the Pareto front. Alternative decoders that explicitly trade off town-switching costs (requiring travel days) against makespan compression could reveal genuine multi-objective trade-offs.
4. **Single-instance testing:** Each configuration uses one instance (seed 42). Testing on multiple instance variants per size would strengthen generalizability claims, especially for varying town-to-location ratios and actor labor limit distributions.

5.10 Key Takeaway

BDM-AOS is not a universally superior algorithm—it underperforms GA/PSO on instances with ≤ 30 scenes. Its value lies in *scalability*: by exploiting the natural community structure of film scheduling via SIG decomposition, it achieves both better solution quality and faster runtime on large instances where scene-level metaheuristics struggle. The 20.7% improvement over GA at 75 scenes and 14.5% at 100 scenes, combined with a $5.5\times$ runtime speedup, demonstrates that decomposition-guided search is the appropriate paradigm for production-scale MSSP instances. The comprehensive 13-constraint formulation with hierarchical location modeling ensures that these improvements are achieved on a realistic and operationally meaningful problem representation.

6 Limitations

Synthetic Data. All instances are synthetically generated following established parameter ranges from the MSSP literature [3, 29, 39]. While our generator incorporates all 13 constraint families—including the three-tier location hierarchy, town logistics, actor labor limits, and criticality weights—it may not capture all idiosyncrasies of real production schedules (e.g., weather dependencies, equipment sharing, creative preferences). Validation on real film scripts, such as the 70-scene instance used by Tekin et al. [39], would strengthen external validity.

Block Granularity. When Louvain decomposition produces too few blocks ($K \leq 4$), the block-permutation search space is too small for evolutionary search to provide value over flat scene-level optimization. On small instances (S10–S15), the BDM variants may underperform scene-level metaheuristics that have a richer search space. Hierarchical decomposition or adaptive block splitting could address this. The ablation results (BDM-Flat vs. BDM-AOS) directly quantify this trade-off.

Pareto Front Collapse. The greedy decoder tends to assign similar makespan values regardless of scene ordering, limiting the diversity of the Pareto front. The bi-objective formulation produces limited cost-makespan trade-offs under the current decoder. A decoder that allows variable daily utilization or explicitly trades off cost against makespan would be needed to realize full multi-objective benefits.

Fixed Hyperparameters. Algorithm hyperparameters (population size, generations, crossover/mutation rates) are fixed across all instance sizes. While our sensitivity analysis (Appendix B) suggests robustness within tested ranges, instance-adaptive parameter tuning could further improve performance, particularly for S75 and S100 where the search landscape changes qualitatively.

Scalability of MILP Baseline. The MILP solver is limited to instances up to S15 within practical time limits (60 seconds), preventing direct comparison with optimal solutions on larger instances. This is a fundamental limitation of exact methods for NP-hard problems [16] and motivates the matheuristic approach, but it means we lack ground-truth optimality gaps for S20 and above.

7 Conclusion

We presented BDM-AOS, a bi-level decomposition matheuristic with adaptive operator selection for the multi-objective Movie Scene Scheduling Problem. Our work makes four primary contributions:

1. A *comprehensive MILP formulation* with 13 constraint families over a three-tier geographic hierarchy (city–town–location), actor–scene–day and location–scene–day linking variables, town-uniqueness and town-continuity logistics constraints, criticality-weighted utilization, and actor consecutive working-day limits—substantially extending prior formulations.
2. The *Scene Interaction Graph* decomposes scenes into shooting blocks based on actor-sharing, location-sharing, and transfer cost structure, reducing the search space from $n!$ to $K!$ permutations. For our largest instance (100 scenes, 24 blocks), this reduction exceeds $10^{134} \times$.
3. The *bi-level framework* combines NSGA-II over block permutations with a global greedy decoder that enforces all 13 constraint families. The decomposition guides scene ordering while cross-block day sharing preserves cost-competitiveness. This is the first decomposition-based approach for the MSSP.
4. *Q-learning adaptive operator selection* with two novel problem-specific operators—Location-Aware Crossover (LAX) and Actor-Continuity Mutation (ACM)—adapts the search strategy based on population diversity, convergence rate, and search progress.

Comprehensive experiments on 8 instance sizes (10–100 scenes) with 30 independent seeds per algorithm-instance pair demonstrate the approach’s scalability. A five-variant ablation study quantifies the contribution of each component: SIG-guided decomposition, adaptive operator selection, and problem-specific operators. The scalability analysis shows that BDM-AOS’s advantages grow with instance size, as the decomposition provides increasingly valuable structure exploitation at larger scales.

7.1 Future Work

Several promising directions emerge:

- **Real-world validation:** Apply BDM-AOS to real film scripts (e.g., the 70-scene, 43-actor instance from Tekin et al. [39]) to validate the hierarchical location model and geographic logistics constraints under real production conditions.
- **Adaptive decomposition:** Re-cluster the SIG periodically during search to exploit evolving population structure, or use hierarchical decomposition (city-level blocks containing town-level sub-blocks) to address the block-granularity limitation.
- **Alternative decoders:** Design decoders that produce genuine cost-makespan trade-offs by allowing variable daily utilization, explicitly trading off town-switching travel days against shooting compactness, and enabling the bi-objective formulation to realize its full potential.
- **Deep RL for AOS:** Replace tabular Q-learning with deep Q-networks for continuous state representation, enabling finer-grained operator selection that generalizes across instance sizes and constraint configurations.

- **Many-objective extension:** Extend to three or more objectives (cost, makespan, actor fatigue, location utilization, town-switching frequency) using NSGA-III [9], leveraging the linking variables $v_{a,s,d}$ and $z_{l,s,d}$ for fine-grained objective computation.
- **Uncertainty handling:** Incorporate stochastic elements (weather delays, actor illness, location unavailability) via robust or stochastic programming at the sub-problem level, using the hierarchical structure to model correlated disruptions (e.g., weather affecting all locations within a town).
- **Extended hierarchy:** Model additional tiers (country-level for international productions) and incorporate travel-day constraints between cities, where town continuity (C12) generalizes to mandatory rest days between city changes.

Reproducibility Statement

We detail the steps taken to ensure full reproducibility of our results.

Code and Data. All source code, synthetic data generators, experiment scripts, and analysis pipelines are publicly available at <https://github.com/VJlaxmi/movie-scheduling-paper>. The repository includes:

- Complete BDM-AOS implementation with all ablation variants (`bdm_aos/` package);
- Baseline implementations (GA, PSO, MILP) using the same evaluation framework;
- Synthetic instance generator with fixed seed (`seed=42`) for deterministic instance creation;
- Experiment runner with 30 independent seeds per algorithm-instance pair;
- Analysis scripts that reproduce all tables and figures from raw JSON results.

Instance Generation. All 8 benchmark instances (S10–S100) are generated deterministically from a single master seed. Instance parameters—including the three-tier location hierarchy (cities, towns, locations), actor criticality weights, location criticality weights, maximum consecutive working days, and hierarchical transfer costs—are documented in Table 2 and Appendix A. The generator produces consistent actor availability calendars, location compatibility matrices, scene-actor assignment matrices, town-location mappings, and all parameters required for the 13 constraint families of Section 3.1.

Algorithm Parameters. All algorithm hyperparameters are fixed throughout the experiments and documented in Section 4. BDM-AOS uses: population size $N = 20$, generations $G = 40$, crossover rate $p_c = 0.9$, mutation rate $p_m = 0.3$, MILP sub-problem time limit 10s. GA uses: population size 50, 100 generations, $p_c = 0.85$, $p_m = 0.15$. PSO uses: swarm size 30, 100 iterations. MILP uses: 60-second time limit with CBC solver.

Statistical Protocol. Each algorithm is run 30 times with seeds 1–30 on each instance. We report mean $\pm$ standard deviation for cost and makespan. Statistical significance is assessed using Friedman tests with Holm’s post-hoc correction at significance level $\alpha = 0.05$. Quality indicators (hypervolume, inverted generational distance) use a common reference point derived from the worst values across all algorithms.

Computational Environment. All experiments were run on a single machine with Python 3.14 and the following key libraries: NumPy, SciPy, pandas, NetworkX, python-louvain (community detection), Pyomo with CBC solver, and Matplotlib. Runtime measurements use Python’s `time.time()` for wall-clock timing.

Broader Impact Statement

Positive Impacts. Efficient movie scene scheduling directly reduces production costs, which can make filmmaking more accessible to independent producers and lower-budget projects. By optimizing actor utilization and minimizing idle days, our approach reduces waste in the form of unnecessary travel, idle crew time, and extended shooting schedules. The explicit modeling of actor consecutive working-day limits (constraint C13) promotes compliance with labor regulations and union rules, ensuring that generated schedules respect worker welfare by construction rather than relying on post-hoc manual correction. The three-tier location hierarchy and town-continuity constraint (C12) produce schedules with coherent geographic progression, reducing disruptive overnight relocations for cast and crew. The decomposition methodology transfers to other resource-constrained scheduling domains including construction project management, operating room scheduling, and manufacturing planning.

Potential Concerns. Automated scheduling tools could reduce human involvement in creative production decisions. Film scheduling involves nuanced factors—actor fatigue, creative flow, weather dependencies—that our cost-centric optimization does not fully capture, despite the criticality weights and labor constraints providing partial modeling of these concerns. We emphasize that BDM-AOS is intended as a decision-support tool that provides optimized candidate schedules for human producers to review and modify, not as a replacement for human judgment.

Environmental Considerations. Optimized scheduling with hierarchical transfer cost minimization directly reduces unnecessary inter-town and inter-city location transfers, which decreases transportation-related carbon emissions. The town-continuity constraint (C12) further reduces emissions by eliminating back-and-forth travel between towns on consecutive days. The computational cost of running BDM-AOS (typically under 10 seconds per instance) is negligible compared to the environmental and financial savings from even marginal schedule improvements.

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A Instance Generation Parameters

Table 8 documents the full parameterization of our synthetic instance generator. All instances are generated with master seed 42 for reproducibility.

Table 8: Synthetic instance generation parameters.

Config	$ S $	$ A $	$ C $	$ T $	$ L $	T
S10	10	5	1	2	3	15
S15	15	8	1	2	4	25
S20	20	10	2	3	5	35
S25	25	12	2	4	6	45
S30	30	15	2	4	7	55
S50	50	20	3	6	10	90
S75	75	35	4	8	12	120
S100	100	43	5	10	15	150

For each instance, parameters are drawn as follows:

Scene parameters.

- Scene durations: $\delta_s \sim U\{1, 2, 3, 4\}$ hours;
- Each scene requires 1–4 actors, assigned with uniform probability;
- Each scene is compatible with 1–2 locations;
- Time windows assigned to $\sim 40\%$ of scenes;
- Soft deadlines assigned to $\sim 30\%$ of scenes, penalties $\rho_s \sim U(50, 500)$.

Actor parameters.

- Daily wages: $w_a \sim U(80, 1000)$ (following Liu et al. [29]);
- Availability: $\sim 80\%$ of the planning horizon;
- Criticality weights: $\kappa_a^A \sim U(1, 10)$;
- Maximum consecutive working days: $\bar{h}_a \sim U\{3, 4, 5, 6\}$.

Location hierarchy parameters.

- Cities: $|\mathcal{C}| = \max(1, \lfloor |\mathcal{L}|/3 \rfloor)$;
- Towns: $|\mathcal{T}| = \max(2, \lfloor 2|\mathcal{L}|/3 \rfloor)$, distributed across cities;
- Location availability: $\sim 85\%$ of the planning horizon;
- Criticality weights: $\kappa_l^L \sim U(1, 10)$;
- Intra-town transfer costs: $\tau^{\text{loc}} \sim U(100, 500)$;
- Inter-town transfer costs: $\tau^{\text{town}} \sim U(500, 2000)$;
- Inter-city transfer costs: $\tau^{\text{city}} \sim U(2000, 8000)$.

Equipment parameters (C14).

- Equipment types: $|\mathcal{E}| = \max(2, \lfloor |\mathcal{L}|/2 \rfloor)$;
- Equipment supply: $\xi_e \sim U\{1, 2\}$ units per type;
- Scene equipment requirements: $\sim 50\%$ of scenes require 1–2 equipment types.

Actor booking parameters (C15).

- Booking windows: $\sim 60\%$ of actors have booking window $[\beta_a^s, \beta_a^e]$, window length $\sim U(T/4, 3T/4)$;
- Holding fees: $h_a \sim U(0.2, 0.6) \times w_a$ (20–60% of daily wage).

Scene coordination parameters (C16–C17).

- Co-scheduling groups: $|\mathcal{G}| = \max(1, \lfloor n/8 \rfloor)$ groups of 2–3 scenes each;
- Separation pairs: $|\mathcal{F}| = \max(1, \lfloor n/10 \rfloor)$ pairs with gap $\delta_{ij} \sim U(2, T/10)$.

Location and lead-actor parameters (C18–C19).

- Location concurrency capacity: $\mu_l \sim U\{1, 2, 3\}$;
- Lead actors: $|\mathcal{A}^P| = \max(1, \lfloor m/5 \rfloor)$ (20% of cast);
- Rest periods: $r_a \sim U\{1, 2\}$ days for $a \in \mathcal{A}^P$.

Structural parameters.

- Precedence constraints: $\sim 20\%$ of scene pairs form a sparse DAG;
- Daily capacity: $W_{\max} = 8$ hours;
- All 19 constraint families (C1–C19) are active.

B Hyperparameter Sensitivity Analysis

We performed a sensitivity analysis of key BDM-AOS hyperparameters on the S25 instance. For each parameter, we varied one parameter while fixing all others at their default values.

Population Size. Varying $N \in \{10, 20, 30, 50\}$: cost improves modestly from $N = 10$ to $N = 20$ (3.2% average reduction) but shows diminishing returns beyond $N = 20$, while runtime scales linearly with N .

Generations. Varying $G \in \{20, 40, 60, 100\}$: convergence is achieved by generation 30–35 for most instances, confirming that $G = 40$ provides sufficient search budget.

Q-Learning Parameters. The AOS learning rate $\alpha \in \{0.05, 0.1, 0.2\}$ and discount factor $\gamma \in \{0.8, 0.9, 0.95\}$ show stable performance across tested ranges, indicating robustness of the Q-learning adaptation.

C Q-Learning State-Action Table

The AOS state space discretizes four continuous features:

- **Generation progress:** early ($g/G < 0.33$), mid ($0.33 \leq g/G < 0.67$), late ($g/G \geq 0.67$);
- **Population diversity:** low ($d < 0.3$), medium ($0.3 \leq d < 0.7$), high ($d \geq 0.7$);
- **Improvement rate:** stagnant ($r < 0.01$), slow ($0.01 \leq r < 0.1$), fast ($r \geq 0.1$);
- **Quality gap:** good ($q < 0.1$), moderate ($0.1 \leq q < 0.5$), poor ($q \geq 0.5$).

This yields $3^4 = 81$ states. The action space comprises $3 \times 3 = 9$ operator pairs: $\{\text{PMX}, \text{OX}, \text{LAX}\} \times \{\text{swap}, \text{block-swap}, \text{ACM}\}$. The Q-table is initialized to zeros and updated via standard Q-learning (Eq. 12).

D Extended Results Tables

Full per-seed results for all 8 algorithms across all 8 instances are provided in the supplementary repository at <https://github.com/VJlaxmi/movie-scheduling-paper/tree/main/experiments/runs>. Each JSON file contains the complete run record including final cost, makespan, runtime, Pareto front, convergence history, and AOS operator frequency log.

E Friedman Test Details

We apply the Friedman test across all algorithms on each instance, using mean cost over 30 seeds as the performance measure. When the Friedman test rejects the null hypothesis of equal ranks ($p < 0.05$), we apply Holm’s step-down procedure for pairwise comparisons with BDM-AOS as the control algorithm.

F Decomposition Analysis

The SIG-based Louvain decomposition produces the following block structures:

Table 9: Block decomposition statistics across instances.

Config	Num. Blocks	Avg. Block Size	Search Space Reduction
S10	3	3.3	$10!/3! = 604,800 \times$
S15	4	3.8	$15!/4! = 5.4 \times 10^{10} \times$
S20	6	3.3	$20!/6! = 3.4 \times 10^{15} \times$
S25	6	4.2	$25!/6! = 2.2 \times 10^{22} \times$
S30	8	3.8	$30!/8! = 8.1 \times 10^{28} \times$
S50	12	4.2	$50!/12! = 1.3 \times 10^{55} \times$
S75	16	4.7	$75!/16! = 5.4 \times 10^{96} \times$
S100	24	4.2	$100!/24! = 4.1 \times 10^{134} \times$

G Runtime Environment

Table 10: Software environment for all experiments.

Component	Version
Python	3.14.2
NumPy	2.3.0
SciPy	1.16.0
pandas	2.3.0
NetworkX	3.5
python-louvain	0.16
Pyomo	6.9.2
CBC Solver	2.10
Matplotlib	3.10.3
OS	Windows